

Analysis — Lecture Notes

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Chapter 1

Sequences

1.1 Convergence

Definition 1.1.1: Convergence

A sequence $(a_n)_{n \in \mathbb{N}}$ in $\mathbb{R}$ converges to $L \in \mathbb{R}$ if, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies |a_n - L| < \varepsilon.$$

Example 1.1.1 (A convergent sequence)

The sequence $a_n = 1/n$ converges to zero. Given $\varepsilon > 0$, choose $N > 1/\varepsilon$; then $n \geq N$ implies $|1/n| \leq 1/N < \varepsilon$.

Theorem 1.1.1 Uniqueness of limits

A convergent real sequence has exactly one limit.

Proof. Suppose $a_n \rightarrow L$ and $a_n \rightarrow M$ with $L \neq M$. Set $\varepsilon = |L - M|/3$. For sufficiently large n ,

$$|L - M| \leq |L - a_n| + |a_n - M| < 2\varepsilon = \frac{2}{3}|L - M|,$$

a contradiction. Thus $L = M$. □

Remark 1.1.1 Notation

Theorem 1.1.1 justifies writing “the limit” of a convergent sequence. Definition 1.1.1 gives its meaning.

1.2 Boundedness

Lemma 1.2.1 Convergent sequences are bounded

Every convergent real sequence is bounded.

Proof. Suppose $a_n \rightarrow L$. Choose N so that $n \geq N$ implies $|a_n - L| < 1$, and set

$$B = \max\{|a_1|, \dots, |a_N|, |L| + 1\}.$$

For $n < N$, the definition of B gives $|a_n| \leq B$. For $n \geq N$, the triangle inequality gives $|a_n| \leq |a_n - L| + |L| < 1 + |L| \leq B$. $\square$

Corollary 1.2.1 A divergence criterion

An unbounded real sequence cannot converge in $\mathbb{R}$.

Note:-

The commands above use section numbering. Use `\thmc`, `\dfnc`, and the other chapter variants for chapter numbering.